Consistent Partial Least Squares: A cSEM tutorial

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Abstract

This tutorial article provides a comprehensive, step-by-step illustration of consistent partial least squares (PLSc) using the cSEM package in R. PLSc is a variant of partial least squares path modeling, a composite-based approach to structural equation modeling, that produces consistent parameter estimates for latent variable models. In particular, PLSc corrects construct correlations for attenuation using Dijkstra-Henseler's composite reliability coefficient ϱ_A. Building on the employee retention example introduced by Cheah, Sarstedt, Hair, and Ringle (2026), we demonstrate how to install and load cSEM, specify the model using lavaan-style syntax, estimate the PLSc-SEM model, evaluate measurement model quality (internal consistency reliability, convergent validity, and discriminant validity), and assess the structural model (collinearity, effect sizes, model fit, and path coefficients with bootstrap inference). We further contrast the PLSc-SEM results with standard PLS-SEM estimates and discuss practical considerations for using PLSc-SEM in empirical research.

Keywords: measurement error, bias correction, partial least squares structural equation modeling (PLS-SEM), factor score regression

1 Introduction

Structural equation modeling (SEM) is a versatile multivariate analysis framework used in the behavioral, social, medical, and management sciences to study complex relationships between constructs and their relations with observed variables (e.g., Kline, 2023; Marcoulides & Schumacker, 1996). To estimate structural equation models, various approaches have been proposed, including maximum-likelihood (Jöreskog, 1970), weighted least squares (Browne, 1974), factor score regression (Devlieger & Rosseel, 2017; Skrondal & Laake, 2001) and (integrated) generalized structured component analysis (Hwang et al., 2021; Hwang & Takane, 2004)

A composite-based estimator to SEM that gained popularity over the last decades in various fields of social sciences (Cook & Forzani, 2023), including marketing (Sarstedt et al., 2022), information systems research (Benitez, Henseler, Castillo, & Schuberth, 2020; Evermann & Rönkkö, 2023), human resource management (Ringle, Sarstedt, Mitchell, & Gudergan, 2020), and tourism research (Müller, Schuberth, & Henseler, 2018), is partial least squares path modeling (PLS-PM, Wold, 1982),also known as partial least squares structural equation modeling (PLS-SEM, Hair, Ringle, & Sarstedt, 2011). PLS-PM is a two-step estimation procedure (Schuberth, Zaza, & Henseler, 2023). In the first step, construct scores are obtained by the partial least squares algorithm. Subsequently, these construct scores are used to estimate the model parameters. Although PLS-PM is computationally efficient, it is known that in its original form to produce inconsistent estimates for models involving latent variables due to attenuation bias (e.g., Dijkstra, 1985; Hui & Wold, 1982; Rönkkö & Evermann, 2013; Schuberth, Rosseel, et al., 2023).

To address this limitation, consistent partial least squares (PLSc, Dijkstra, 2013; Dijkstra & Henseler, 2015a, 2015b; Dijkstra & Schermelleh-Engel, 2014) was introduced. PLSc is based on PLS-PM, but applies a correction for attenuation to obtain consistent estimates of reflective measurement models and path coefficient between latent variables. Additionally, PLSc has been extended to deal with ordinal variables (Schuberth, Henseler, & Dijkstra, 2018), randomly occurring outliers (Schamberger, Schuberth, Henseler, & Dijkstra, 2020) and multicollinearity issues (Jung & Park, 2018).

To apply PLS-PM and PLSc, various implementations have been developed (Venturini, Mehmetoglu, & Latan, 2023). They can be distinguished into: (i) proprietary software and (ii) open-source software. Proprietary software for conducting PLS-PM and PLSc includes SmartPLS (Ringle, Wende, & Becker, 2024), WarpPLS (Kock, 2025) and ADANCO (Henseler, 2025). On the other hand, open-source software alternatives include implementations in the statistical programming environment R (R Core Team, 2025) such as matrixpls (Rönkkö, 2022), SEMinR (Ray, Danks, & Calero Valdez, 2026), cSEM (Rademaker & Schuberth, 2020) and plspm (Sanchez, Trinchera, Russolillo, & Bertrand, 2025). Similarly, it includes implementations in python (Python Software Foundation, 2024) such as plspm (Humble, 2024). While proprietary software such as SmartPLS often shows high ease of use (Cheah, Magno, & Cassia, 2024), they conflict with the open science principles (**?**). For example, their codebase is not openly accessible of transparency (Morin et al., 2012), reproducibility (Ince, Hatton, & Graham-Cumming, 2012), and accessibility/inclusiveness (). Open-source software overcomes these limitations (Barba, 2022). Therefore, this tutorial presents the R package cSEM as a full-fledged open-source alternative to SmartPLS to apply PLS-PM and PLSc.

Barba (2022)

(Chuah et al., 2021)

The remainder of the tutorial is structured as follows. Section Section Section

2 cSEM R Illustration

2.1 Getting started with R

To use the cSEM package, users first need to install R and an integrated development environment (IDE) such as RStudio. For the installation, we refer to the official guides for R (`https://cran.r-project.org`) and RStudio (`https://posit.co/download/rstudio-desktop`).

When RStudio is opened for the first time, it displays four panes (Figure 1). The two most important ones are the *script editor* (upper left), where code is written and saved, and the *Console* (lower left), where the code is executed and results are printed. The *Environment* pane (upper right) lists the objects currently loaded into memory, and the fourth pane (lower right) provides access to files, plots, and help pages.

Although code can be typed directly into the console, it is good practice to work in a script, since a script can be saved, shared, and re-run, which is essential for reproducible analyses. To create a new script, click File → New File → R Script. The typical workflow is to write code in the editor and run the selected line(s) with `Ctrl + Enter`. Any line beginning with `#` is treated as a comment: it documents what the code does and is ignored when the code runs.

```
# This is a comment and is not executed.
1 + 1    # Code is executed; the result is printed in the Console.

[1] 2
```

Before cSEM can be used, it must be installed. The cSEM package is available on CRAN, so the most recent stable version can be installed with:

```
install.packages("cSEM")
```

Installation only needs to be done once per machine (rerun this line to update the package later). Moreover, development versions can be installed from GitHub: `https://github.com/FloSchuberth/cSEM`. After installation, the package must be loaded at the start of every new R session before its functions become available:

```
library(cSEM)


Attache Paket:  'cSEM'
Das folgende Objekt ist maskiert 'package:stats':
      predict
Das folgende Objekt ist maskiert 'package:grDevices':
      savePlot
```

With cSEM installed and loaded, we can now import the data and specify the model, as described in the following sections.

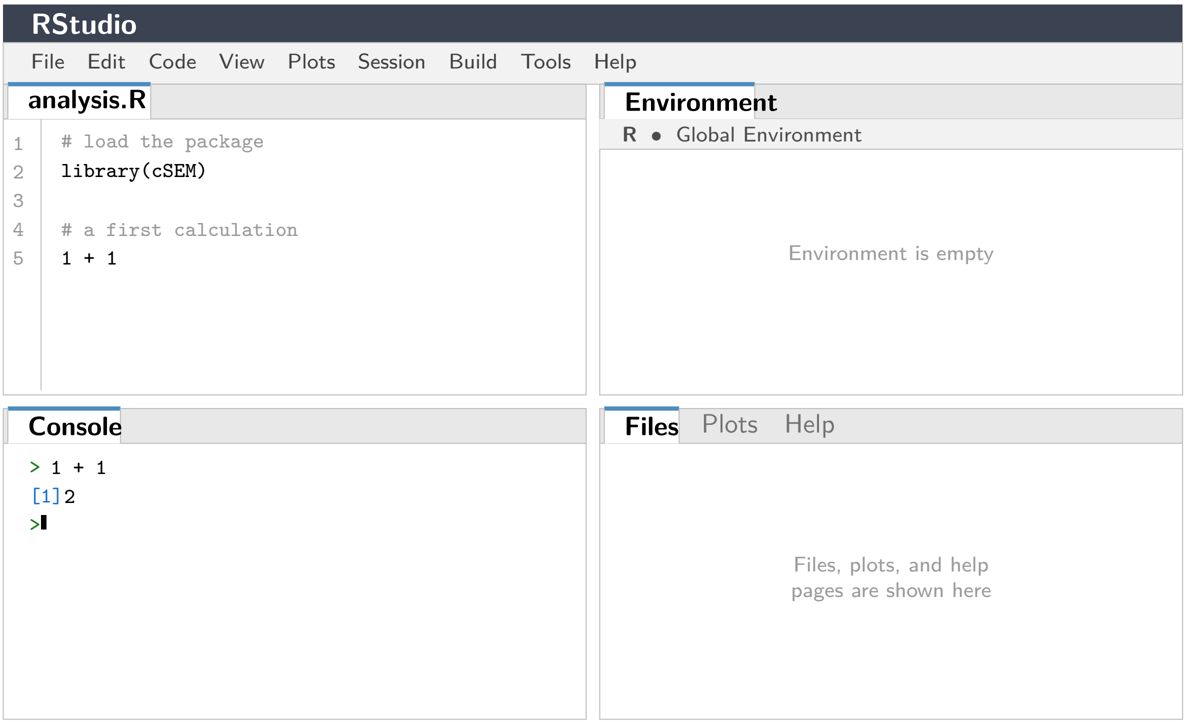

Figure 1: The four panes of the RStudio interface: the script editor (upper left), the *Console* (lower left), the *Environment* pane (upper right), and the files/plots/help pane (lower right).

2.2 Model and Data

The cSEM package (Rademaker & Schuberth, 2020) offers a variety of composite-based approaches to SEM including PLS-PM and PLSc. To illustrate the application of PLSc using cSEM, we use the data and model used in Cheah et al. (2026) to demonstrate SmartPLS. Similarly, we compare the results to those of the maximum-likelihood estimation as implemented in the R package lavaan (Rosseel, 2012; Rosseel, Jorgensen, & De Wilde, 2025).

Following Cheah et al. (2026), our illustration draws on the employee retention model introduced by Hair, Black, Babin, and Anderson (2019, Chapter 13). The model explains the effects of organizational commitment (OC) and job satisfaction (JS) on employees' staying intentions (SI), with work environment perceptions (EP) and attitudes toward coworkers (AC) as exogenous antecedent constructs. Five constructs are reflectively measured by a total of 21 indicators. A detailed description of the indicators can be found in Cheah et al. (2026). The model is presented in Figure 2.

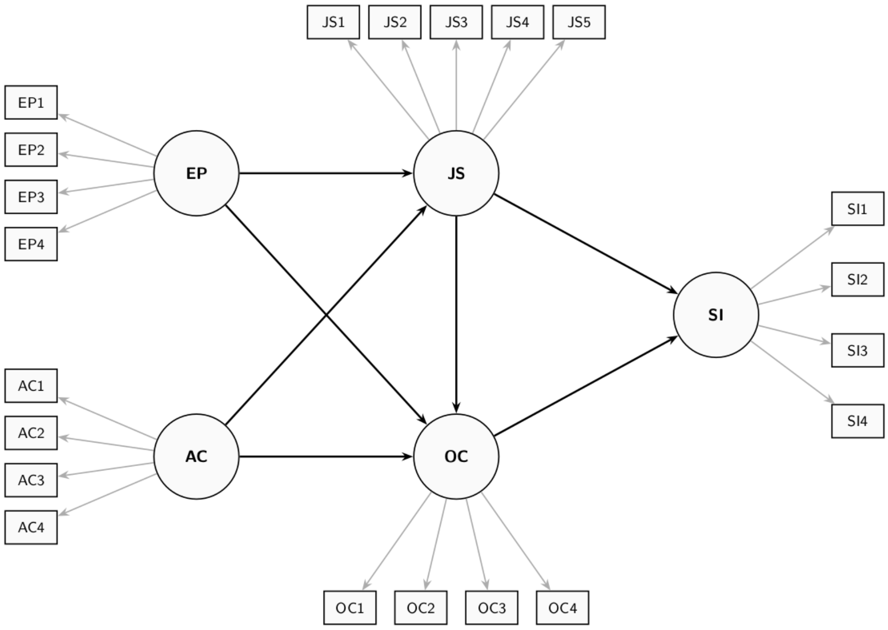

Figure 2: Employee Retention Model

The dataset used for model estimation is publicly available from the SmartPLS website: `https://www.smartpls.com/documentation/sample-projects/employee-retention`. The synthetic dataset comprises $N = 400$ observations from HBAT Industries and was generated to satisfy the assumptions of factor-based SEM, including multivariate normality. For this tutorial, we assume the data have been downloaded as a CSV file. After downloading the dataset, it needs to be loaded into the R Environment. The Employee Retention dataset is stored as an CSV File using ';' as the separator, thus it can be loaded into R using following command:

```
# Import dataset
HBAT_SEM <- read.csv("data/HBAT_SEM.csv", sep=";")

# Select relevant variables
HBAT_SEM_rel <- HBAT_SEM[, c("AC1", "AC2", "AC3", "AC4",
                   "EP1", "EP2", "EP3", "EP4",
                   "JS1", "JS2", "JS3", "JS4", "JS5",
                   "OC1", "OC2", "OC3", "OC4",
                   "SI1", "SI2", "SI3", "SI4")]

# show dimensions of the relevant dataset
dim(HBAT_SEM_rel)

[1] 400  21

# show some basic statistics for all variables in the dataset
summary(HBAT_SEM_rel)

      AC1            AC2             AC3             AC4       
 Min.   :1.00   Min.   :1.000   Min.   :1.000   Min.   :1.000  
 1st Qu.:1.75   1st Qu.:2.000   1st Qu.:1.000   1st Qu.:2.000  
 Median :3.00   Median :4.000   Median :3.000   Median :3.000  
 Mean   :2.76   Mean   :3.553   Mean   :2.775   Mean   :3.211  
 3rd Qu.:4.00   3rd Qu.:5.000   3rd Qu.:4.000   3rd Qu.:4.000  
```

```
 Max.   :5.00   Max.   :6.000   Max.   :5.000   Max.   :6.000  
                                                NA's   :1      
      EP1              EP2              EP3              EP4      
 Min.   : 0.000   Min.   : 0.000   Min.   : 0.000   Min.   :1.00  
 1st Qu.: 8.000   1st Qu.: 8.000   1st Qu.: 8.000   1st Qu.:5.00  
 Median : 9.000   Median : 9.000   Median : 9.000   Median :6.00  
 Mean   : 8.527   Mean   : 8.848   Mean   : 8.928   Mean   :5.83  
 3rd Qu.:10.000   3rd Qu.:10.000   3rd Qu.:10.000   3rd Qu.:7.00  
 Max.   :10.000   Max.   :10.000   Max.   :10.000   Max.   :7.00  
                                                    NA's   :1     
      JS1             JS2             JS3             JS4       
 Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000  
 1st Qu.:3.000   1st Qu.:3.000   1st Qu.:2.000   1st Qu.:2.000  
 Median :4.000   Median :4.000   Median :3.000   Median :3.000  
 Mean   :4.197   Mean   :4.202   Mean   :3.215   Mean   :2.667  
 3rd Qu.:5.000   3rd Qu.:5.000   3rd Qu.:4.000   3rd Qu.:4.000  
 Max.   :7.000   Max.   :7.000   Max.   :6.000   Max.   :5.000  

      JS5               OC1              OC2              OC3        
 Min.   :  0.00   Min.   : 0.000   Min.   : 0.000   Min.   : 0.000  
 1st Qu.: 39.97   1st Qu.: 3.000   1st Qu.: 8.000   1st Qu.: 8.000  
 Median : 55.35   Median : 5.000   Median : 9.000   Median : 9.000  
 Mean   : 54.83   Mean   : 4.888   Mean   : 8.418   Mean   : 8.655  
 3rd Qu.: 70.44   3rd Qu.: 7.000   3rd Qu.:10.000   3rd Qu.:10.000  
 Max.   :100.00   Max.   :10.000   Max.   :10.000   Max.   :10.000  

      OC4              SI1             SI2             SI3       
 Min.   : 0.000   Min.   :1.000   Min.   :1.000   Min.   :1.000  
 1st Qu.: 8.000   1st Qu.:4.000   1st Qu.:4.000   1st Qu.:3.000  
 Median : 9.000   Median :4.000   Median :4.000   Median :3.000  
 Mean   : 8.405   Mean   :4.202   Mean   :4.205   Mean   :3.473  
 3rd Qu.:10.000   3rd Qu.:5.000   3rd Qu.:5.000   3rd Qu.:4.000  
 Max.   :10.000   Max.   :5.000   Max.   :5.000   Max.   :5.000  

      SI4      
 Min.   :1.00  
 1st Qu.:3.00  
 Median :4.00  
 Mean   :3.48  
 3rd Qu.:4.00  
 Max.   :5.00  
```

In the summary, we see that we have missing values in the variables AC4 and EP4. Before we can continue we need to use inputation methods or delete incomplete observations. Deleting incomplete observations can be done via following command:

```
# filter the dataset, only select complete cases
HBAT_SEM_rel <- HBAT_SEM_rel[complete.cases(HBAT_SEM_rel),]

#  the number of rows changed from 400 to 398
dim(HBAT_SEM_rel)

[1] 398  21
```

When using their own data, users must ensure it is stored as a data frame or a matrix before passing it to cSEM functions; if necessary, the data can be converted with `as.data.frame()` or `as.matrix()`.

2.3 Model specification

In cSEM, models are specified using the lavaan model syntax. The lavaan package does not need to be installed to use the syntax. In particular, the `=~` and the `<~` operators are used to define the relations between the constructs and their indicators. Specifically, the `=~` operator is used to specify reflective measurement models, while the `<~` is used to specify composite models (see , for an explanation of reflective and composite models). In addition, the `~` operator is used to specify the relations among constructs. In case, an observed variable should be directly accounted for in the structural model, it must be specified as a single indicator construct, as common in PLS-PM (). The cSEM specification of the retention model illustrated in Figure 2 is as follows:

```
model <- "
  # Measurement model
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
  SI =~ SI1 + SI2 + SI3 + SI4

  # Structural model
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"
```

3 Estimating the model with cSEM

3.1 Running the estimation

The central function is `csem()`. The minimal input required for the csem fucntion is a dataset and a model:

.data A data frame or matrix containing the observed indicator variables.

.model A model string written in lavaan syntax, with $= /sim$ for measurement relations and $\sim$ for structural (regression) relations.

To ultimately estimate with the csem function, we call it and save the results as

```
# PLSc-SEM estimation
res_csem <- csem(
  .data  = HBAT_SEM_rel,
  .model = model
)

# Verify convergence
verify(res_csem)
```

```
---------------------------------------------------------------------------------

Verify admissibility:

  admissible

Details:

  Code   Status    Description
  1      ok        Convergence achieved
  2      ok        All absolute standardized loading estimates <= 1
  3      ok        Construct VCV is positive semi-definite
  4      ok        All reliability estimates <= 1
  5      ok        Model-implied indicator VCV is positive semi-definite
---------------------------------------------------------------------------------
```

The `verify()` function checks whether the PLS algorithm converged, whether all construct reliability estimates (ρ_A) are positive (a necessary condition for the PLSc correction), and whether the model-implied indicator correlation matrix is admissible. If all checks pass, the estimation can be considered successful.

To obtain a comprehensive overview of all results, including loadings, path coefficients, reliability measures, and R^2 values, use:

```r
# Full results summary
summary_res <- summarize(res_csem)
print(summary_res)
```

```
---------------------------------------------------------------------------------
----------------------------------- Overview -----------------------------------

 General information:
 ------------------------
 Estimation status                  = Ok
 Number of observations             = 398
 Weight estimator                   = PLS-PM
 Inner weighting scheme             = "path"
 Type of indicator correlation      = Pearson
 Path model estimator               = OLS
 Second-order approach              = NA
 Type of path model                 = Linear
 Disattenuated                      = Yes (PLSc)

 Construct details:
 -------------------
 Name  Modeled as     Order         Mode

 EP    Common factor  First order   "modeA"
 AC    Common factor  First order   "modeA"
 JS    Common factor  First order   "modeA"
 OC    Common factor  First order   "modeA"
 SI    Common factor  First order   "modeA"

----------------------------------- Estimates -----------------------------------
```

```
Estimated path coefficients:
============================
  Path       Estimate  Std. error   t-stat.   p-value
  JS ~ EP      0.2448          NA        NA        NA
  JS ~ AC     -0.0033          NA        NA        NA
  OC ~ EP      0.4288          NA        NA        NA
  OC ~ AC      0.1726          NA        NA        NA
  OC ~ JS      0.0961          NA        NA        NA
  SI ~ JS      0.1312          NA        NA        NA
  SI ~ OC      0.4900          NA        NA        NA

Estimated loadings:
===================
  Loading      Estimate  Std. error   t-stat.   p-value
  EP =~ EP1      0.6341          NA        NA        NA
  EP =~ EP2      0.8787          NA        NA        NA
  EP =~ EP3      0.7678          NA        NA        NA
  EP =~ EP4      0.8231          NA        NA        NA
  AC =~ AC1      0.7691          NA        NA        NA
  AC =~ AC2      0.6765          NA        NA        NA
  AC =~ AC3      0.8216          NA        NA        NA
  AC =~ AC4      0.9934          NA        NA        NA
  JS =~ JS1      0.6248          NA        NA        NA
  JS =~ JS2      0.6965          NA        NA        NA
  JS =~ JS3      0.7182          NA        NA        NA
  JS =~ JS4      0.6319          NA        NA        NA
  JS =~ JS5      0.9004          NA        NA        NA
  OC =~ OC1      0.4252          NA        NA        NA
  OC =~ OC2      0.9575          NA        NA        NA
  OC =~ OC3      0.6859          NA        NA        NA
  OC =~ OC4      0.8566          NA        NA        NA
  SI =~ SI1      0.8170          NA        NA        NA
  SI =~ SI2      0.8705          NA        NA        NA
  SI =~ SI3      0.6778          NA        NA        NA
  SI =~ SI4      0.8959          NA        NA        NA

Estimated weights:
==================
  Weight       Estimate  Std. error   t-stat.   p-value
  EP <~ EP1      0.2425          NA        NA        NA
  EP <~ EP2      0.3361          NA        NA        NA
  EP <~ EP3      0.2937          NA        NA        NA
  EP <~ EP4      0.3148          NA        NA        NA
  AC <~ AC1      0.2707          NA        NA        NA
  AC <~ AC2      0.2381          NA        NA        NA
  AC <~ AC3      0.2892          NA        NA        NA
  AC <~ AC4      0.3496          NA        NA        NA
  JS <~ JS1      0.2223          NA        NA        NA
  JS <~ JS2      0.2478          NA        NA        NA
  JS <~ JS3      0.2556          NA        NA        NA
  JS <~ JS4      0.2249          NA        NA        NA
  JS <~ JS5      0.3204          NA        NA        NA
```

```
  OC <~ OC1      0.1741          NA        NA        NA
  OC <~ OC2      0.3920          NA        NA        NA
  OC <~ OC3      0.2808          NA        NA        NA
  OC <~ OC4      0.3507          NA        NA        NA
  SI <~ SI1      0.2882          NA        NA        NA
  SI <~ SI2      0.3071          NA        NA        NA
  SI <~ SI3      0.2391          NA        NA        NA
  SI <~ SI4      0.3161          NA        NA        NA

Estimated construct correlations:
=================================
  Correlation    Estimate  Std. error   t-stat.   p-value
  EP ~~ AC         0.2549          NA        NA        NA

------------------------------------ Effects -----------------------------------

Estimated total effects:
========================
  Total effect    Estimate  Std. error   t-stat.   p-value
  JS ~ EP           0.2448          NA        NA        NA
  JS ~ AC          -0.0033          NA        NA        NA
  OC ~ EP           0.4523          NA        NA        NA
  OC ~ AC           0.1722          NA        NA        NA
  OC ~ JS           0.0961          NA        NA        NA
  SI ~ EP           0.2538          NA        NA        NA
  SI ~ AC           0.0840          NA        NA        NA
  SI ~ JS           0.1783          NA        NA        NA
  SI ~ OC           0.4900          NA        NA        NA

Estimated indirect effects:
===========================
  Indirect effect    Estimate  Std. error   t-stat.   p-value
  OC ~ EP              0.0235          NA        NA        NA
  OC ~ AC             -0.0003          NA        NA        NA
  SI ~ EP              0.2538          NA        NA        NA
  SI ~ AC              0.0840          NA        NA        NA
  SI ~ JS              0.0471          NA        NA        NA
________________________________________________________________________________

# For a more compact overview
summary_res$Estimates$Loading_estimates

        Name Construct_type  Estimate Std_err t_stat p_value
1  EP =~ EP1  Common factor 0.6341011      NA     NA      NA
2  EP =~ EP2  Common factor 0.8786819      NA     NA      NA
3  EP =~ EP3  Common factor 0.7678021      NA     NA      NA
4  EP =~ EP4  Common factor 0.8230691      NA     NA      NA
5  AC =~ AC1  Common factor 0.7691299      NA     NA      NA
6  AC =~ AC2  Common factor 0.6765259      NA     NA      NA
7  AC =~ AC3  Common factor 0.8215860      NA     NA      NA
8  AC =~ AC4  Common factor 0.9933858      NA     NA      NA
9  JS =~ JS1  Common factor 0.6248337      NA     NA      NA
10 JS =~ JS2  Common factor 0.6964587      NA     NA      NA
11 JS =~ JS3  Common factor 0.7182274      NA     NA      NA
```

```
12 JS =~ JS4  Common factor 0.6319045      NA      NA       NA
13 JS =~ JS5  Common factor 0.9004003      NA      NA       NA
14 OC =~ OC1  Common factor 0.4252105      NA      NA       NA
15 OC =~ OC2  Common factor 0.9574682      NA      NA       NA
16 OC =~ OC3  Common factor 0.6858765      NA      NA       NA
17 OC =~ OC4  Common factor 0.8565670      NA      NA       NA
18 SI =~ SI1  Common factor 0.8170099      NA      NA       NA
19 SI =~ SI2  Common factor 0.8705194      NA      NA       NA
20 SI =~ SI3  Common factor 0.6778349      NA      NA       NA
21 SI =~ SI4  Common factor 0.8959341      NA      NA       NA

summary_res$Estimates$Path_estimates

     Name Construct_type     Estimate Std_err t_stat p_value
1 JS ~ EP   Common factor  0.244824960      NA      NA       NA
2 JS ~ AC   Common factor -0.003282073      NA      NA       NA
3 OC ~ EP   Common factor  0.428781485      NA      NA       NA
4 OC ~ AC   Common factor  0.172554413      NA      NA       NA
5 OC ~ JS   Common factor  0.096069429      NA      NA       NA
6 SI ~ JS   Common factor  0.131226588      NA      NA       NA
7 SI ~ OC   Common factor  0.490013776      NA      NA       NA
```

3.2 Assessing the measurement model

Following the guidelines described in Hair, Hult, Ringle, and Sarstedt (2017, Chapter 4) and Sarstedt, Ringle, and Hair (n.d.), we assess the reflective measurement model in terms of internal consistency reliability, convergent validity, and discriminant validity. The `assess()` function in cSEM provides all relevant quality criteria:

```
# Compute all quality criteria
quality <- assess(res_csem)

Warning:  The following warning occured in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning:  The following warning occured in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning in log(x[[3]]):  NaNs wurden erzeugt

# Internal consistency reliability
quality$Cronbachs_alpha      # Cronbach's alpha

NULL

quality$RhoA                 # Dijkstra-Henseler rho_A

NULL

quality$RhoC                 # Composite reliability (Joreskog rho_C)

       EP        AC        JS        OC        SI 
0.8607098 0.8918737 0.8417418 0.8343858 0.8901518 
```

```
# Convergent validity
quality$AVE                  # Average variance extracted

       EP        AC        JS        OC        SI 
0.6102822 0.6777668 0.5202693 0.5754208 0.6718668 

# Indicator loadings
summary_res$Estimates$Loading_estimates

        Name Construct_type  Estimate Std_err t_stat p_value
1  EP =~ EP1  Common factor 0.6341011      NA     NA      NA
2  EP =~ EP2  Common factor 0.8786819      NA     NA      NA
3  EP =~ EP3  Common factor 0.7678021      NA     NA      NA
4  EP =~ EP4  Common factor 0.8230691      NA     NA      NA
5  AC =~ AC1  Common factor 0.7691299      NA     NA      NA
6  AC =~ AC2  Common factor 0.6765259      NA     NA      NA
7  AC =~ AC3  Common factor 0.8215860      NA     NA      NA
8  AC =~ AC4  Common factor 0.9933858      NA     NA      NA
9  JS =~ JS1  Common factor 0.6248337      NA     NA      NA
10 JS =~ JS2  Common factor 0.6964587      NA     NA      NA
11 JS =~ JS3  Common factor 0.7182274      NA     NA      NA
12 JS =~ JS4  Common factor 0.6319045      NA     NA      NA
13 JS =~ JS5  Common factor 0.9004003      NA     NA      NA
14 OC =~ OC1  Common factor 0.4252105      NA     NA      NA
15 OC =~ OC2  Common factor 0.9574682      NA     NA      NA
16 OC =~ OC3  Common factor 0.6858765      NA     NA      NA
17 OC =~ OC4  Common factor 0.8565670      NA     NA      NA
18 SI =~ SI1  Common factor 0.8170099      NA     NA      NA
19 SI =~ SI2  Common factor 0.8705194      NA     NA      NA
20 SI =~ SI3  Common factor 0.6778349      NA     NA      NA
21 SI =~ SI4  Common factor 0.8959341      NA     NA      NA
```

The results in Table **??** indicate that all constructs meet the commonly accepted thresholds. All Cronbach's α and ρ_{A} values exceed 0.70, and all AVE values exceed 0.50, supporting internal consistency reliability and convergent validity. While some outer loadings (e.g., OC1 $= 0.427$) fall below the recommended threshold of 0.708, these indicators are retained because the corresponding constructs exhibit adequate reliability and AVE, and their retention preserves content validity (**?**).

3.2.1 Discriminant Validity: The HTMT Criterion

Discriminant validity is assessed using the heterotrait-monotrait ratio of correlations (HTMT; Henseler, Ringle, & Sarstedt, 2015). In `cSEM`, HTMT values are available via `assess()`:

```
# HTMT values
quality$HTMT

$htmts
          EP         AC        JS        OC SI
EP 1.0000000 0.00000000 0.0000000 0.0000000  0
AC 0.2555939 1.00000000 0.0000000 0.0000000  0
JS 0.2437979 0.05239146 1.0000000 0.0000000  0
OC 0.4939933 0.27449892 0.2086788 1.0000000  0
```

```
SI 0.5672796 0.30936763 0.2313363 0.5005239  1

$quantiles
NULL

$nr_admissibles
NULL

# Bootstrap confidence intervals for HTMT
res_plsc_boot <- csem(
  .data              = HBAT_SEM_rel,
  .model             = model,
  .approach_weights = "PLS-PM",
  .disattenuate      = TRUE,
  .resample_method   = "bootstrap",
  .R                 = 10000,
  .seed              = 42
)

Warning:  Paket 'future' wurde unter R Version 4.5.3 erstellt

# Extract HTMT inference
htmt_infer <- infer(res_plsc_boot)
# The infer() output includes CIs for all quality criteria
# including HTMT values
```

Table 1: HTMT criterion results.

Construct	AC	EP	JS	OC
EP	0.257 [0.163; 0.347]			
JS	0.066 [0.032; 0.075]	0.244 [0.164; 0.326]		
OC	0.275 [0.195; 0.355]	0.495 [0.405; 0.580]	0.209 [0.118; 0.297]	
SI	0.310 [0.222; 0.395]	0.569 [0.471; 0.656]	0.232 [0.154; 0.313]	0.501 [0.410; 0.586]

Note. Values in brackets: 90% percentile bootstrap confidence intervals (10,000 subsamples).

The HTMT results in Table 1 confirm discriminant validity: none of the upper bounds of the 90% bootstrap confidence intervals exceed the conservative threshold of 0.85 (Franke & Sarstedt, 2019; Henseler et al., 2015). These findings are identical to those reported by Cheah et al. (2026, Table 2), confirming that the cSEM results replicate the SmartPLS output.

3.3 Assessing the structural model

3.3.1 Collinearity Assessment

Before interpreting the structural model, we check for collinearity among predictor constructs by examining variance inflation factors (VIF). In cSEM, VIF values are available through the `assess()` function:

```
# VIF values for structural model
quality$VIF
```

```
          EP       AC       JS       OC
JS 1.069517 1.069517 0.000000 0.000000
OC 1.133251 1.069528 1.063310 0.000000
SI 0.000000 0.000000 1.046545 1.046545
```

All VIF values should be below the critical threshold of three (**?**), indicating that multicollinearity is not a concern.

3.3.2 Path Coefficients and Bootstrap Inference

To evaluate the statistical significance of the structural relationships, we rely on bootstrap confidence intervals. The bootstrap estimation was already conducted above. Path coefficients and their confidence intervals are extracted as follows:

```
# Extract path coefficient estimates
summarize(res_plsc)$Estimates$Path_estimates

Error:
! Objekt 'res_plsc' nicht gefunden

# Bootstrap inference for path coefficients
boot_infer <- infer(res_plsc_boot)

# Percentile bootstrap confidence intervals (95%)
boot_infer$Path_estimates$CI_percentile

       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC       OC ~ JS    SI ~ JS
95%L 0.1525445 -0.1247503 0.3226357 0.09125872 -0.004465693 0.03809319
95%U 0.3584825  0.1147835 0.5483708 0.27329485  0.200425662 0.22330878
       SI ~ OC
95%L 0.3987664
95%U 0.5977819
```

Table 2: Structural model assessment results.

Relationship	Path coeff.	SD	t value	p value	95% CI	VIF	f^2
AC → JS	−0.002	0.058	0.043	.966	[−0.119; 0.106]	1.055	0.000
AC → OC	0.172	0.048	3.619	.000	[0.078; 0.264]	1.055	0.039
EP → JS	0.245	0.053	4.641	.000	[0.136; 0.343]	1.055	0.060
EP → OC	0.430	0.059	7.272	.000	[0.309; 0.540]	1.101	0.227
JS → OC	0.095	0.052	1.820	.069	[−0.011; 0.194]	1.047	0.012
JS → SI	0.132	0.046	2.833	.005	[0.043; 0.225]	1.035	0.023
OC → SI	0.490	0.051	9.520	.000	[0.386; 0.587]	1.035	0.321

Note. CI = 95% percentile bootstrap confidence interval (10,000 subsamples). Results replicate Cheah et al. (2026, Table 3).

The structural model results in Table 2 are consistent with those reported by Cheah et al. (2026, Table 3). Based on the 95% percentile bootstrap confidence intervals, all structural relationships are statistically significant except for the paths from AC to JS ($\beta = -0.002$, $p = .966$) and from JS to OC ($\beta = 0.095$, $p = .069$). Environment perceptions (EP) exert the strongest positive effect on organizational commitment ($\beta = 0.430$), which in turn is the dominant predictor of staying intentions ($\beta = 0.490$).

3.3.3 Effect Sizes

Cohen's f^2 effect sizes quantify the substantive impact of each predictor. In cSEM:

```
# Effect sizes
quality$F2

           EP           AC          JS         OC SI
JS 0.05959141 1.070949e-05 0.00000000 0.0000000  0
OC 0.22615825 3.880845e-02 0.01209977 0.0000000  0
SI 0.00000000 0.000000e+00 0.02299583 0.3206432  0
```

The paths EP → OC ($f^2 = 0.227$) and OC → SI ($f^2 = 0.321$) show medium to large effects, while the remaining significant paths exhibit small effects. The AC → JS relationship has a trivial effect size of essentially zero.

3.3.4 Model Fit Assessment

Following Schuberth, Rosseel, et al. (2023), we assess model fit using the `testOMF()` function, which implements a bootstrap-based test comparing the empirical and model-implied correlation matrices:

```
# Overall model fit test
fit_test <- testOMF(
  res_plsc,
  .R                        = 10000,
  .seed                     = 42,
  .handle_inadmissibles     = "replace"
)

Error:
!  Objekt 'res_plsc' nicht gefunden

# Print results
fit_test

Error:
!  Objekt 'fit_test' nicht gefunden

# The output includes:
# - SRMR and its bootstrap-based critical value
# - dULS (squared Euclidean distance) and critical value
# - dG (geodesic distance) and critical value
```

Table 3: Model fit statistics for PLSc-SEM estimation.

	Saturated model	Estimated model
SRMR	0.044	0.071
d_{ULS}	0.438; $\mathrm{UB}_{95\%}$: 0.636	1.155; $\mathrm{UB}_{95\%}$: 0.641
d_{G}	0.303; $\mathrm{UB}_{95\%}$: 0.953	0.330; $\mathrm{UB}_{95\%}$: 0.870

Note. $\mathrm{UB}_{95\%}$ = 95th percentile of the bootstrap distribution (10,000 subsamples).

Consistent with Cheah et al. (2026, Table 4), the SRMR of 0.071 falls below the conventional 0.08 threshold (Hu & Bentler, 1999), supporting adequate model fit. The bootstrap-based test yields mixed results: d_{ULS} exceeds its critical value (rejecting fit), while d_{G} falls below its critical value (supporting fit). Overall, the model exhibits a satisfactory level of fit.

4 Customizing the PLS algorithm

Beyond the data and the model, `csem()` exposes a number of optional arguments. The defaults follow the recommendations established in (), so the minimal call already returns appropriate estimates. Users who wish to adjust the PLS algorithm can do so through a set of dedicated arguments, which govern the weighting scheme, the convergence behaviour, and the iteration process:

4.1 Customization options

`.conv_criterion` Character string specifying the criterion used to check for convergence of the weighting algorithm; one of `"diff_absolute"`, `"diff_squared"`, or `"diff_relative"`, with `"diff_absolute"` as the default.

`.disattenuate` Logical indicating whether composite (proxy) correlations as well as loadings and path estimates should be disattenuated to yield consistent estimates when at least one construct is modeled as a common factor; defaults to `TRUE`.

`.dominant_indicators` A character vector of `"construct_name" = "indicator_name"` pairs naming the dominant indicator for a (subset of) construct(s), which fixes the sign/orientation of the corresponding composite; defaults to `NULL`.

`.id` A character string or integer giving the name or column position of the variable in `.data` whose levels are used to split the data into groups for a multigroup analysis; defaults to `NULL`.

`.iter_max` Integer giving the maximum number of iterations allowed for the weighting algorithm; with `.iter_max = 1` and `.approach_weights = "PLS-PM"` one-step weights are returned, and exceeding the maximum returns the weights of the last step with a warning (default `100`).

`.normality` Logical indicating whether joint normality of the latent variables and error terms should be assumed when estimating a nonlinear model; defaults to `FALSE` and is ignored for linear models.

`.PLS_approach_cf` Character string selecting the approach used to obtain the correction factors for PLSc (e.g. `"dist_squared_euclid"`, `"dist_euclid_weighted"`, `"fisher_transformed"`, the various means, or `"geo_of_harmonic"`); defaults to `"dist_squared_euclid"` and is ignored unless disattenuation is applied with PLS-PM.

`.PLS_ignore_structural_model` Logical indicating whether the structural model should be ignored when computing the inner weights of the PLS-PM algorithm; defaults to `FALSE` and is ignored for non-PLS-PM weighting approaches.

`.PLS_modes` Specifies the mode used to estimate the weights of each construct in PLS-PM (e.g. `"modeA"`, `"modeB"`, `"modeBNNLS"`, `"unit"`, `"PCA"`, or user-supplied fixed weights), either as a named list per construct or a single value for all; defaults to `NULL`, in which case the appropriate mode is chosen automatically from the construct type.

.PLS_weight_scheme_inner Character string giving the inner weighting scheme used by PLS-PM; one of `"centroid"`, `"factorial"`, or `"path"`, with `"path"` as the default (ignored for non-PLS-PM approaches).

.reliabilities A character vector of `"name" = value` pairs supplying known reliabilities (between 0 and 1) for a (subset of) construct(s); defaults to `NULL`, in which case reliabilities are estimated by `csem()` (currently only supported for `.approach_weights = "PLS-PM"`).

.starting_values A named list of vectors of `"indicator_name" = value` pairs giving the (scaled or unscaled) starting indicator weights for the weighting algorithm of selected constructs; defaults to `NULL`.

.tolerance Double giving the tolerance criterion against which the convergence of the weighting algorithm is judged; defaults to `1e-05`.

4.2 Comparison of different customizations

An instructive exercise is to compare PLSc-SEM with standard PLS-SEM to see the effect of the consistency correction. In cSEM, this is achieved simply by toggling `.disattenuate`:

```
# Standard PLS-SEM (no correction)
res_pls <- csem(
  .data              = hbat_sub,
  .model             = model,
  .approach_weights  = "PLS-PM",
  .disattenuate      = FALSE
)

Error:
!  Objekt 'hbat_sub' nicht gefunden

# Side-by-side comparison
plsc_est <- summarize(res_plsc)$Estimates$Path_estimates

Error:
!  Objekt 'res_plsc' nicht gefunden

pls_est  <- summarize(res_pls)$Estimates$Path_estimates

Error:
!  Objekt 'res_pls' nicht gefunden

comparison <- merge(
  plsc_est[, c("Name", "Estimate")],
  pls_est[, c("Name", "Estimate")],
  by = "Name", suffixes = c("_PLSc", "_PLS")
)

Error:
!  Objekt 'plsc_est' nicht gefunden

print(comparison)

Error:
!  Objekt 'comparison' nicht gefunden
```

As expected from the methodological literature (Dijkstra & Henseler, 2015a), the standard PLS-SEM path coefficients are attenuated (biased toward zero) relative to the PLSc-SEM estimates. The attenuation is most visible for paths involving constructs with lower reliability (e.g., OC, which has a relatively low loading for OC1). The PLSc correction recovers the consistent estimates that would be expected under the common factor model.

5 Comparing PLS with alternative estimators

A key advantage of working in R is the seamless ability to compare PLSc-SEM results with those from CB-SEM using the lavaan package (Rosseel, 2012). This allows researchers to conduct the multimethod estimation strategy advocated by Cheah et al. (2026) and Sarstedt et al. (2024) entirely within one script:

```
# CB-SEM via lavaan
library(lavaan)

This is lavaan 0.6-21
lavaan is FREE software!  Please report any bugs.

lavaan_model <- "
  # Measurement model
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
  SI =~ SI1 + SI2 + SI3 + SI4

  # Structural model
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"

fit_cbsem <- sem(lavaan_model, data = hbat_sub)

Error:
!  Objekt 'hbat_sub' nicht gefunden

summary(fit_cbsem, fit.measures = TRUE, standardized = TRUE)

Error in `h()`:
!  Fehler bei der Auswertung des Argumentes 'object' bei der Methodenauswahl fÃ¼r Funktion
'summary':  Objekt 'fit_cbsem' nicht gefunden

# Extract standardized path coefficients
parameterEstimates(fit_cbsem, standardized = TRUE) |>
  subset(op == "~", select = c(lhs, op, rhs, est, std.all, pvalue))

Error:
!  Objekt 'fit_cbsem' nicht gefunden
```

The lavaan model specification is identical to the cSEM specification, making side-by-side comparison straightforward. We can extract and compare the structural path coefficients:

```
# Compare structural path coefficients
# PLSc-SEM
plsc_paths <- summarize(res_plsc)$Estimates$Path_estimates

Error:
! Objekt 'res_plsc' nicht gefunden

# CB-SEM
cbsem_paths <- parameterEstimates(fit_cbsem, standardized = TRUE)

Error:
! Objekt 'fit_cbsem' nicht gefunden

cbsem_struct <- cbsem_paths[cbsem_paths$op == "~",
                            c("lhs", "rhs", "std.all")]

Error:
! Objekt 'cbsem_paths' nicht gefunden

# Model fit comparison
fitMeasures(fit_cbsem, c("chisq", "df", "pvalue", "rmsea",
                         "rmsea.ci.lower", "rmsea.ci.upper",
                         "srmr", "cfi", "tli", "nfi"))

Error in `h()`:
! Fehler bei der Auswertung des Argumentes 'object' bei der Methodenauswahl fÃ¼r Funktion
'fitMeasures': Objekt 'fit_cbsem' nicht gefunden
```

In line with the findings of Cheah et al. (2026, Figure 4), the CB-SEM and PLSc-SEM path coefficient estimates closely correspond. The CB-SEM model fit indices (RMSEA = 0.038, SRMR = 0.060, CFI = 0.976) confirm good model fit, supporting the conclusions drawn from the PLSc-SEM analysis.

6 Further cSEM functionality

7 Discussion

This tutorial has demonstrated how to conduct a complete PLSc-SEM analysis using the cSEM package in R, replicating the employee retention model analysis presented by Cheah et al. (2026) using SmartPLS 4. The results obtained via cSEM are identical to those reported in the SmartPLS-based tutorial, confirming that both software implementations produce consistent estimates.

The cSEM package offers several advantages for researchers seeking a script-based, open-source approach to PLSc-SEM. First, the use of R scripts ensures full reproducibility: every step from data import to final inference is documented in executable code that can be shared, reviewed, and re-run. Second, working within the R ecosystem provides seamless integration with CB-SEM via lavaan, enabling the multimethod estimation strategy advocated by Cheah et al. (2026) and Sarstedt et al. (2024) within a single analytical pipeline. Third, cSEM provides a rich set of postestimation tools—including `testOMF()` for model fit, `predict()` for out-of-sample prediction, and `testMICOM()` for measurement invariance—that support a comprehensive model evaluation workflow.

Several practical considerations warrant mention. Like SmartPLS, cSEM allows toggling between standard PLS-SEM and PLSc-SEM with a single argument (`.disattenuate`), making it straightforward to assess the impact of the consistency correction. Furthermore, cSEM supports additional weighting approaches beyond PLS-PM, including generalized structured component analysis (GSCA) weights and various regression-based approaches, offering flexibility for sensitivity analyses.

However, users should be aware of certain limitations. The PLSc correction requires that all ρ_A estimates are positive and that the resulting corrected correlation matrix is positive definite. The `verify()` function checks these conditions, but researchers should inspect warning messages carefully, especially with small samples or weakly measured constructs. As Cheah et al. (2026) note, the attenuation correction in PLSc-SEM alters the original PLS-SEM path coefficients, which means that predictive assessment methods such as $\text{PLS}_{\text{predict}}$ (Shmueli et al., 2019) are not directly applicable in the PLSc framework.

We hope this tutorial serves as a practical guide for researchers who wish to leverage the strengths of PLSc-SEM within an open-source, reproducible analytical framework. Coupled with the SmartPLS-based tutorial by Cheah et al. (2026), researchers now have comprehensive guidance for conducting PLSc-SEM analyses across both GUI-based and script-based software environments.

A A compact reproducible workflow

For convenience, we present a compact, self-contained R script that reproduces the complete PLSc-SEM analysis. This script can serve as a template for researchers applying PLSc-SEM with cSEM to their own datasets:

```
# ============================================================
# PLSc-SEM Analysis with cSEM: Compact Reproducible Script
# ============================================================

# 1. Setup ----
library(cSEM)

hbat <- read.csv("HBAT_Employee_Retention.csv", sep = ";")

Warning in file(file, "rt"):  kann Datei 'HBAT_Employee_Retention.csv' nicht Ã¶ffnen:
No such file or directory
Error in 'file()':
!  kann Verbindung nicht Ã¶ffnen

vars <- c(paste0("AC", 1:4), paste0("EP", 1:4),
          paste0("JS", 1:5), paste0("OC", 1:4),
          paste0("SI", 1:4))
dat <- hbat[, vars]

Error:
!  Objekt 'hbat' nicht gefunden

# 2. Model specification ----
model <- "
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
```

```
  SI =~ SI1 + SI2 + SI3 + SI4
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"

# 3. PLSc-SEM estimation ----
res <- csem(dat, model,
            .approach_weights = "PLS-PM",
            .disattenuate = TRUE)
```

```
Error:
! Objekt 'dat' nicht gefunden
```

```
verify(res)
```

```
Error:
! Objekt 'res' nicht gefunden
```

```
# 4. Assessment ----
q <- assess(res)
```

```
Error:
! Objekt 'res' nicht gefunden
```

```
q$Cronbachs_alpha; q$RhoA; q$RhoC; q$AVE
```

```
Error in `q$Cronbachs_alpha`:
! Objekt des Typs 'closure' ist nicht indizierbar
```

```
q$HTMT; q$VIF; q$F2
```

```
Error in `q$HTMT`:
! Objekt des Typs 'closure' ist nicht indizierbar
```

```
# 5. Bootstrap inference ----
res_boot <- csem(dat, model,
                 .approach_weights = "PLS-PM",
                 .disattenuate = TRUE,
                 .resample_method = "bootstrap",
                 .R = 10000, .seed = 42)
```

```
Error:
! Objekt 'dat' nicht gefunden
```

```
infer(res_boot)
```

```
Error:
! Objekt 'res_boot' nicht gefunden
```

```
# 6. Model fit ----
testOMF(res, .R = 10000, .seed = 42,
        .handle_inadmissibles = "replace")
```

```
Error:
! Objekt 'res' nicht gefunden
```

References

Barba, L. A. (2022). Defining the role of open source software in research reproducibility. *Computer*, *55*(8), 40–48. doi: 10.1109/mc.2022.3177133

Benitez, J., Henseler, J., Castillo, A., & Schuberth, F. (2020). How to perform and report an impactful analysis using partial least squares: Guidelines for confirmatory and explanatory IS research. *Information & Management*, *2*(57), 103168. doi: 10.1016/j.im.2019.05.003

Browne, M. W. (1974). Generalized least squares estimators in the analysis of covariance structures. *South African Statistical Journal*, *8*(1), 1–24. doi: 10.1002/j.2333-8504.1973.tb00197.x

Cheah, J.-H., Magno, F., & Cassia, F. (2024). Reviewing the SmartPLS 4 software: the latest features and enhancements. *Journal of Marketing Analytics*, *12*, 97–102. doi: 10.1057/s41270-023-00266-y

Cheah, J.-H., Sarstedt, M., Hair, J. F., & Ringle, C. M. (2026). Consistent partial least squares structural equation modeling using smartpls. *Structural Equation Modeling: A Multidisciplinary Journal*, 1–14. doi: 10.1080/10705511.2026.2633754

Chuah, F., Memon, M. A., Thurasamy, R., Cheah, J.-H., Ting, H., & Cham, T. H. (2021, aug). PLS-SEM using R: An introduction to cSEM and SEMinR. *Journal of Applied Structural Equation Modeling*, *5*(2), 1–35. doi: 10.47263/jasem.5(2)01

Cook, R. D., & Forzani, L. (2023). On the role of partial least squares in path analysis for the social sciences. *Journal of Business Research*, *167*, 114132. doi: 10.1016/j.jbusres.2023.114132

Devlieger, I., & Rosseel, Y. (2017). Factor score path analysis. *Methodology*, *13*, 31-38.

Dijkstra, T. K. (1985). *Latent variables in linear stochastic models: Reflections on "maximum likelihood" and "partial least squares" methods* (Vol. 1). Amsterdam: Sociometric Research Foundation.

Dijkstra, T. K. (2013). *A note on how to make pls consistent* (Tech. Rep.). working paper, University of Groningen, Groningen, The Netherlands. Retrieved from `http://www.rug.nl/staff/tkdijkstra/how-to-make-pls-consistent.pdf`

Dijkstra, T. K., & Henseler, J. (2015a). Consistent and asymptotically normal PLS estimators for linear structural equations. *Computational Statistics & Data Analysis*, *81*, 10–23. doi: 10.1016/j.csda.2014.07.008

Dijkstra, T. K., & Henseler, J. (2015b). Consistent partial least squares path modeling. *MIS Quarterly*, *39*(2), 297–316. doi: 10.25300/MISQ/2015/39.2.02

Dijkstra, T. K., & Schermelleh-Engel, K. (2014). Consistent partial least squares for nonlinear structural equation models. *Psychometrika*, *79*(4), 585–604. doi: 10.1007/s11336-013-9370-0

Evermann, J., & Rönkkö, M. (2023). Recent developments in PLS. *Communications of the Association for Information Systems*, *52*, 663–667. doi: 10.17705/1CAIS.05229

Franke, G., & Sarstedt, M. (2019). Heuristics versus statistics in discriminant validity testing: a comparison of four procedures. *Internet Research*, *29*(3), 430–447. doi: 10.1108/IntR-12-2017-0515

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). *Multivariate data analysis* (8th ed.). Cengage.

Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2017). *A primer on partial least squares structural equation modeling (PLS-SEM)* (Second edition ed.). SAGE.

Hair, J. F., Ringle, C. M., & Sarstedt, M. (2011). PLS-SEM: Indeed a silver bullet. *Journal of Marketing Theory and Practice*, *19*(2), 139–152. doi: 10.2753/MTP1069-6679190202

Henseler, J. (2025). Adanco 2.4 [Computer software manual]. Enschede.

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, *43*(1), 115–135. doi: 10.1007/s11747-014-0403-8

Hu, L.-t., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, *6*(1), 1–55. doi: 10.1080/10705519909540118

Hui, B. S., & Wold, H. (1982). Consistency and consistency at large of partial least squares estimates. In K. G. Jöreskog & H. Wold (Eds.), *Systems under indirect observation: Causality, structure, prediction part ii* (pp. 119–130). Amsterdam: North-Holland.

Humble, J. (2024). plspm: A library implementing partial least squares path modeling [Computer software manual]. Retrieved from `https://plspm.readthedocs.io/` (Python package version 0.5.7)

Hwang, H., Cho, G., Jung, K., Falk, C., Flake, J., Jin, M. J., & Lee, S. H. (2021). An approach to structural equation modeling with both factors and components: Integrated generalized structured component analysis. *Psychological Methods*, *26*(3), 273–294. doi: 10.1037/met0000336

Hwang, H., & Takane, Y. (2004). Generalized structured component analysis. *Psychometrika*, *69*(1), 81–99. doi: 10.1007/BF02295841

Ince, D. C., Hatton, L., & Graham-Cumming, J. (2012). The case for open computer programs. *Nature*, *482*(7386), 485–488. doi: 10.1038/nature10836

Jöreskog, K. G. (1970). A general method for estimating a linear structural equation system. *ETS Research Bulletin Series*, *1970*(2), i–41. doi: 10.1002/j.2333-8504.1970.tb00783.x

Jung, S., & Park, J. (2018). Consistent partial least squares path modeling via regularization. *Frontiers in Psychology*, *9*. doi: 10.3389/fpsyg.2018.00174

Kline, R. B. (2023). *Principles and practice of structural equation modeling* (5th ed.). New York: The Guilford Press.

Kock, N. (2025). *WarpPLS user manual: Version 8.0.* Laredo, TX: ScriptWarp Systems.

Marcoulides, G. A., & Schumacker, R. E. (Eds.). (1996). *Advanced structural equation modeling: Issues and techniques.* New York: Psychology Press.

Morin, A., Urban, J., Adams, P. D., Foster, I., Sali, A., Baker, D., & Sliz, P. (2012). Shining light into black boxes. *Science*, *336*(6078), 159–160. doi: 10.1126/science.1218263

Müller, T., Schuberth, F., & Henseler, J. (2018). PLS path modeling - a confirmatory approach to study tourism technology and tourist behavior. *Journal of Hospitality and Tourism Technology*, *9*(3), 249–266. doi: 10.1108/JHTT-09-2017-0106

Python Software Foundation. (2024). Python language reference, version 3.12 [Computer software manual]. Retrieved from `https://www.python.org`

R Core Team. (2025). R: A language and environment for statistical computing [Computer software manual]. Vienna, Austria. Retrieved from `https://www.R-project.org/`

Rademaker, M. E., & Schuberth, F. (2020). csem: Composite-based structural equation modeling [Computer software manual]. Retrieved from `https://floschuberth.github.io/cSEM/` (Package version: 0.6.1.9000)

Ray, S., Danks, N. P., & Calero Valdez, A. (2026). seminr: Building and estimating structural equation models [Computer software manual]. Retrieved from `https://CRAN.R-project.org/package=seminr` (R package version 2.4.2) doi: 10.32614/CRAN.package.seminr

Ringle, C. M., Sarstedt, M., Mitchell, R., & Gudergan, S. P. (2020, jan). Partial least squares structural equation modeling in HRM research. *The International Journal of Human Resource Management*, *31*(12), 1617–1643. doi: 10.1080/09585192.2017.1416655

Ringle, C. M., Wende, S., & Becker, J.-M. (2024). *SmartPLS 4.* Boenningstedt: SmartPLS GmbH. Retrieved from `https://www.smartpls.com`

Rönkkö, M. (2022). matrixpls: Matrix-based partial least squares estimation [Computer software manual]. Retrieved from `https://github.com/mronkko/matrixpls` (R package version 1.0.15)

Rönkkö, M., & Evermann, J. (2013). A critical examination of common beliefs about partial least squares path modeling. *Organizational Research Methods*, *16*(13), 425–448. doi: 10.1177/1094428112474693

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2), 1–36. doi: 10.18637/jss.v048.i02

Rosseel, Y., Jorgensen, T. D., & De Wilde, L. (2025). lavaan: Latent variable analysis [Computer software manual]. Retrieved from `https://CRAN.R-project.org/package=lavaan` (R package version 0.6-20) doi: 10.32614/CRAN.package.lavaan

Sanchez, G., Trinchera, L., Russolillo, G., & Bertrand, F. (2025). Partial least squares path modeli (PLS-PM, Frederic) [Computer software manual]. Retrieved from `https://CRAN.R-project.org/package=plspm` (R package version 0.6.0) doi: 10.32614/CRAN.package.plspm

Sarstedt, M., Adler, S. J., Ringle, C. M., Cho, G., Diamantopoulos, A., Hwang, H., & Liengaard, B. D. (2024). Same model, same data, but different outcomes: Evaluating the impact of method choices in structural equation modeling. *Journal of Product Innovation Management*, *41*(6), 1100–1117. doi: 10.1111/jpim.12738

Sarstedt, M., Hair, J. F., Pick, M., Liengaard, B. D., Radomir, L., & Ringle, C. M. (2022). Progress in partial least squares structural equation modeling use in marketing research in the last decade. *Psychology & Marketing*, *39*, 1035–1064. doi: 10.1002/mar.21640

Sarstedt, M., Ringle, C. M., & Hair, J. F. (n.d.). Partial least squares structural equation modeling. In *Handbook of market research* (pp. 587–632). Springer, Cham. Retrieved 2026-06-17, from `https://link.springer.com/rwe/10.1007/978-3-319-57413-4_15` doi: 10.1007/978-3-319-57413-4_15

Schamberger, T., Schuberth, F., Henseler, J., & Dijkstra, T. K. (2020). Robust partial least squares path modeling. *Behaviormetrika*, *47*, 307–334. doi: 10.1007/s41237-019-00088-2

Schuberth, F., Henseler, J., & Dijkstra, T. K. (2018). Partial least squares path modeling using ordinal categorical indicators. *Quality & Quantity*, *52*(1), 9–35. doi: 10.1007/s11135-016-0401-7

Schuberth, F., Rosseel, Y., Rönkkö, M., Trinchera, L., Kline, R. B., & Henseler, J. (2023). Structural parameters under partial least squares and covariance-based structural equation modeling: A comment on Yuan and Deng (2021). *Structural Equation Modeling: A Multidisciplinary Journal*, *30*(3), 339–345. doi: 10.1080/10705511.2022.2134140

Schuberth, F., Zaza, S., & Henseler, J. (2023). Partial least squares is an estimator for structural equation models: A comment on Evermann and Rönkkö (2021). *Communications of the Association for Information Systems*, *52*, 711–729. doi: 10.17705/1CAIS.05232

Shmueli, G., Sarstedt, M., Hair, J. F., Cheah, J.-H., Ting, H., Vaithilingam, S., & Ringle, C. M. (2019). Predictive model assessment in PLS-SEM: guidelines for using PLSpredict. *European Journal of Marketing*, *53*(11), 2322–2347. doi: 10.1108/EJM-02-2019-0189

Skrondal, A., & Laake, P. (2001). Regression among factor scores. *Psychometrika*, *66*(4), 563–575. doi: 10.1007/BF02296196

Venturini, S., Mehmetoglu, M., & Latan, H. (2023). Software packages forÂ partial least squares structural equation modeling: An updated review. In *Partial least squares path modeling* (pp. 113–152). Springer International Publishing. doi: 10.1007/978-3-031-37772-3_5

Wold, H. (1982). Soft modeling: The basic design and some extensions. In K. G. Jöreskog & H. Wold (Eds.), *Systems under indirect observation: Causality, structure, prediction part ii* (pp. 1–54). Amsterdam: North-Holland.